Module - IV

Fourier & Z-transform

Integral Transform

Integral transform of $f(p)$ of a function $F(x)$ is defined as,

$$I\{F(x)\} = f(p) = \int_a^b F(x)\ K(p,x)\ dx$$

where, $K(p,x)$ is a function of p & x, called Kernel of transform. P is called parameter and $F(x)$ is inverse transform of $f(p)$.

Complex Fourier Transform

Let $K = e^{ipx}$ then

$$F\{F(x)\} = f(p) = \int_{-\infty}^{\infty} F(x)\ e^{ipx}\ dx$$

$F(x)$ is a function defined for $-\infty \leq x \leq \infty$.

Fourier Integral

$$F(x) = \frac{1}{\pi} \int_0^{\infty} \int_{-\infty}^{\infty} F(t) \cdot \cos \lambda(t-x)\, dt\, d\lambda$$

This equation is called the Fourier Integral of $F(x)$.

At a point of discontinuity,

$$F(x) = \frac{1}{2}\left[F(x+0) + F(x-0)\right].$$

Fourier Sine Integral

$$F_S(x) = \frac{2}{\pi} \int_0^{\infty} \sin \lambda x \left(\int_0^{\infty} F(t) \sin \lambda t\, dt \right) d\lambda$$

Fourier Cosine Integral

$$F_C(x) = \frac{2}{\pi} \int_0^{\infty} \cos \lambda x \left(\int_0^{\infty} F(t) \cos \lambda t\, dt \right) d\lambda$$

V. Imp.

Q.1) Express the function $F(x) = \begin{cases} 1, & |x| \le 1 \\ 0, & |x| > 1 \end{cases}$ as a fourier integral. Hence evaluate:

$$\int_0^\infty \frac{\sin\lambda \cdot \cos\lambda x}{\lambda}\, d\lambda$$

Sol^n:

$$F(x) = \begin{cases} 1, & -1 \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$$

Fourier integral representation,

$$F(x) = \frac{1}{\pi}\int_0^\infty \left[\int_{-\infty}^{\infty} f(t)\cos\lambda(t-x)\,dt\right] d\lambda$$

$[F(t) = 1 \text{ for } -1 \le t \le 1]$

$$\Rightarrow F(x) = \frac{1}{\pi}\int_0^\infty \left[\int_{-1}^{1} \cos\lambda(t-x)\,dt\right] d\lambda$$

$$= \frac{1}{\pi}\int_0^\infty \left[\frac{\sin\lambda(t-x)}{\lambda}\right]_{-1}^{1} d\lambda$$

$$= \frac{1}{\pi}\int_0^\infty \left[\frac{\sin\lambda(1-x)}{\lambda} - \frac{\sin\lambda(-1-x)}{\lambda}\right] d\lambda$$

$$= \frac{1}{\pi}\int_0^\infty \left[\frac{\sin\lambda(1-x) + \sin\lambda(1+x)}{\lambda}\right] d\lambda$$

$$\left[\sin A + \sin B = 2\sin\left(\frac{A+B}{2}\right)\cdot\cos\left(\frac{A-B}{2}\right)\right]$$

$$\Rightarrow F(x) = \frac{2}{\pi}\int_0^\infty \frac{\sin\left(\frac{x\lambda}{2}\right)\cdot\cos\left(\frac{-x\lambda x}{2}\right)}{\lambda}\, d\lambda$$

$$\Rightarrow F(x) = \frac{2}{\pi}\int_0^\infty \frac{\sin\lambda\cdot\cos\lambda x}{\lambda}\, d\lambda$$

This is the required representation.

$$\frac{\pi}{2}F(x) = \int_0^\infty \frac{\sin\lambda\cos\lambda x}{\lambda}\, d\lambda$$

For $|x| < 1$,

$$\int_0^\infty \frac{\sin\lambda\cos\lambda x}{\lambda}\, d\lambda = \frac{\pi}{2}\cdot 1 = \frac{\pi}{2}$$

For $|x| > 1$,

$$\int_0^\infty \frac{\sin\lambda\cos\lambda x}{\lambda}\, d\lambda = \frac{\pi}{2}\cdot 0 = 0$$

For $|x| = 1$, $\frac{F(x^+) + F(x^-)}{2} = \frac{1+0}{2} = \frac{1}{2}$

$$\int_0^\infty \frac{\sin\lambda\cos\lambda x}{\lambda}\, d\lambda = \frac{\pi}{2}\cdot\frac{1}{2} = \frac{\pi}{4}$$

$$\therefore \int_0^\infty \frac{\sin\lambda\cos\lambda x}{\lambda}\, d\lambda = \begin{cases} \pi/2, & |x| < 1 \\ \pi/4, & |x| = 1 \\ 0, & |x| > 1 \end{cases}$$

$$\int e^{at} \sin(bt)\, dt = \frac{e^{at}}{a^2+b^2}(a\sin(bt) - b\cos(bt))$$

Q.2) Using Fourier integral representation, show that

$$\int_0^\infty \frac{\omega \sin x\omega}{1+\omega^2} = \frac{\pi}{2} e^{-x}, \quad x > 0$$

Soln:

$$F(x) = e^{-x}$$

Sine integral of e^{-x},

$$e^{-x} = \frac{2}{\pi}\int_0^\infty \sin \lambda x \left(\int_0^\infty e^{-t} \sin \lambda t\, dt\right) d\lambda$$

$$\Rightarrow e^{-x} = \frac{2}{\pi}\int_0^\infty \sin \lambda x \left[\frac{e^{-t}}{1+\lambda^2}(-\sin \lambda t - \lambda \cos \lambda t)\right]_0^\infty d\lambda$$

$$\Rightarrow e^{-x} = \frac{2}{\pi}\int_0^\infty \sin \lambda x \left[\frac{-1}{1+\lambda^2}\left(\frac{-\sin 0}{0} - \lambda \frac{\cos 0}{1}\right)\right] d\lambda$$

$$\Rightarrow e^{-x} = \frac{2}{\pi}\int_0^\infty \frac{\lambda \sin \lambda x}{1+\lambda^2}\, d\lambda$$

$$\Rightarrow \int_0^\infty \frac{\lambda \sin \lambda x}{1+\lambda^2}\, d\lambda = \frac{\pi}{2} e^{-x}$$

i.e. $$\int_0^\infty \frac{\omega \sin \omega x}{1+\omega^2}\, dx = \frac{\pi}{2} e^{-x}$$ proved

$$\int e^{ax}\cos(bx)\,dx = \frac{e^{ax}}{a^2+b^2}\cdot(a\cos(bx)+b\sin(bx))$$

Q.3) Using Fourier integral configuration, prove that:

$$\int_0^\infty \frac{\cos\omega x}{1+\omega^2} = \frac{\pi}{2}e^{-x}.$$

Sol^n:

$$F(x) = e^{-x}$$

Cosine integral of e^{-x} is,

$$e^{-x} = \frac{2}{\pi}\int_0^\infty \cos\lambda x\left(\int_0^\infty e^{-x}\cos\lambda t\,dt\right)d\lambda$$

$$\Rightarrow e^{-x} = \frac{2}{\pi}\int_0^\infty \cos\lambda x\left[\frac{e^{-x}}{1+\lambda^2}(-\cos\lambda t+\lambda\sin\lambda t)\right]_0^\infty d\lambda$$

$$\Rightarrow e^{-x} = \frac{2}{\pi}\int_0^\infty \cos\lambda x\left[0-\frac{1}{1+\lambda^2}\left(\frac{-\cos 0}{1}+\frac{\lambda\sin 0}{0}\right)\right]d\lambda$$

$$\Rightarrow e^{-x} = \frac{2}{\pi}\int_0^\infty \cos\lambda x\left[\frac{+1}{1+\lambda^2}\right]d\lambda$$

$$\Rightarrow \int_0^\infty \frac{\cos\lambda x}{1+\lambda^2}\,d\lambda = \frac{\pi}{2}e^{-x}$$

i.e. $$\int_0^\infty \frac{\cos\omega x}{1+\omega^2} = \frac{\pi}{2}e^{-x}$$ proved

Complex Fourier Transform

$$F\{F(x)\} = f(p) = \int_{-\infty}^{\infty} F(x) \cdot e^{ipx}\, dx$$

Inverse -

$$F^{-1}(f(p)) = F(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(p) \cdot e^{-ipx}\, dx$$

Fourier Sine Transform

$$F_s\{F(x)\} = f_s(p) = \int_0^{\infty} F(x) \sin px\, dx$$

Inverse, $F_s^{-1}(f_s(p)) = \frac{2}{\pi} \int_0^{\infty} f_s(p) \sin px\, dp$

Fourier Cosine Transform

$$F_c\{F(x)\} = f_c(p) = \int_0^{\infty} F(x) \cos px\, dx$$

Inverse, $F_c^{-1}(f_c(p)) = \frac{2}{\pi} \int_0^{\infty} f_c(p) \cos px\, dp$.

Properties of Fourier Transform

1. Linearly Property

$$F(C_1F_1(x) + C_2F_2(x)) = C_1F(F_1(x)) + C_2F(F_2(x))$$

2. ~~Shifting~~ Change of Scale Property

$$F(F(ax)) = \frac{1}{|a|} f\left(\frac{\rho}{a}\right)$$

3. Shifting Property

$$F(F(x-a)) = e^{\pm iap} f(\rho)$$

Modulation Theorem

If $f(\rho)$ is complex fourier transform of $F(x)$, then,

$$F(F(x)\cos(ax)) = \frac{1}{2}[f(\rho+a) + f(\rho-a)]$$

Convolution Theorem

Convolution of two function $F(x)$ & $G(x)$ is,

$$F*G(x) = \int_{-\infty}^{\infty} F(u).G(x-u)\,du$$

& Fourier transform of above is,

$$F(F*G) = F(F(x)).F(G(x)) = f(\rho).g(\rho).$$